A Report on

AI Resume & Job Match Analyzer

carried out as part of the course Emerging Technologies (Project)

Submitted by

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BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

Under the Guidance of:

Project Guide

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CERTIFICATE

This is to certify that the project entitled "AI Resume & Job Match Analyzer" is a Bonafide work carried out as part of Emerging Technologies Project in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by Anirudh jain (23FE10CSE00772) during the academic semester of year 2026.

Place: Manipal University Jaipur

Name of the project guide: Dr. Jai Prakash Singh

Signature of the project guide: _____

Date: _____

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We dedicate this project to technical innovation and practical problem-solving in the domain of resume analysis and career guidance.

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ABSTRACT

This project presents an AI-assisted Resume and Job Match Analyzer built using FastAPI and a lightweight web interface. The system evaluates resume quality, recommends suitable roles, and compares resume-job fit through ATS-style scoring. For ML Engineer roles specifically, it provides role-targeted skill gap analysis, actionable ATS suggestions, and rewritten bullet points emphasizing machine learning impact.

The system employs a sophisticated four-layer matching pipeline — exact matching, strict fuzzy matching (cutoff 0.75), medium fuzzy matching (cutoff 0.60), and a transparent not-found response — ensuring accurate recommendations without false matches. A resume-upgrade module generates an improved version with before-vs-after comparison and produces an Overleaf-ready LaTeX draft with PDF preview support.

The system supports multiple analysis modes: OpenAI API, Groq API, local Ollama, or deterministic fallback heuristic analyzer. To eliminate per-query model rebuilding overhead, all similarity matrices are computed once at application startup and held persistently in memory, achieving sub-100ms query response after initial loading.

The system is deployed as an interactive web application with a professional interface. Comprehensive testing across representative test cases demonstrates reliable recommendation quality with correct skill-based matching. The solution is designed for accessibility, remaining functional without paid APIs by supporting local and heuristic analysis modes, making it ideal for students and early-career professionals.

Keywords: Resume Analysis, ATS Optimization, Content-Based Filtering, FastAPI, ML Engineer, Skill Matching, Career Guidance

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1. Introduction

1.1 Objective of the Project

The objective of this project is to build and deploy an AI-assisted Resume and Job Match Analyzer that operates across multiple technical roles and job domains. Using a sophisticated analysis engine with multiple backend support (OpenAI, Groq, Local LLM, or deterministic heuristic), the system computes resume-job fit through ATS-style scoring. The system aims to: (i) score CV readiness using role-agnostic heuristics, (ii) suggest relevant roles based on resume content, (iii) perform role-specific matching with targeted skill overlap analysis, and (iv) generate an upgraded resume draft with before-vs-after comparison highlighting improvements.

1.2 Problem Statement

Digital content platforms and job marketplaces host thousands of job descriptions. Manual resume review is time-consuming, inconsistent, and often misses critical ATS (Applicant Tracking System) keywords. For specialized technical roles like ML Engineer, identifying skill gaps and optimizing resumes manually requires significant effort. Existing tools either cost money, require external APIs, or lack multi-role support. REWIND solves this by applying content-based filtering and intelligent skill matching to provide affordable, accessible, role-targeted resume analysis without mandatory paid subscriptions.

1.3 Significance of the Work

Resume optimization is critical for career progression, especially in competitive technical domains. This project demonstrates complete applied understanding of unsupervised machine learning, text processing, skill extraction, and web deployment in a real-world context. By implementing end-to-end from raw dataset preprocessing through feature engineering, similarity computation, and cloud deployment, the project is both technically substantive and practically demonstrable. The multi-role scope across diverse positions and the graceful fallback logic from paid APIs to free heuristics addresses a gap in student-level ML projects.

2. Literature Review

2.1 Presentation of the Problem

Resume screening and role matching have been studied in information retrieval and recommendation systems literature for decades. The core challenge is constructing a meaningful similarity metric that captures semantic overlap between resume content and job requirements without conflating unrelated similarities. Traditional approaches rely on keyword matching, but modern systems employ TF-IDF vectorization, embeddings, and semantic similarity metrics. For specialized roles like ML Engineer, the discriminative features (TensorFlow, PyTorch, MLOps) differ from general software engineering roles, requiring domain-aware feature engineering.

2.2 Existing Approaches and Gaps

Reference	Method	Key Finding	Gap Addressed
[1]	TF-IDF + Cosine Similarity	Effective for document retrieval	Original foundation adapted for resume matching
[]	Content-Based Filtering	Genre/features predict item similarity	Skill-based feature engineering applied
[]	Fuzzy String Matching	Handles typos and near-matches	Multi-layer pipeline prevents false positives
[]	LLM-Based Analysis	Semantic understanding of text	Graceful fallback to deterministic methods

Key gaps: Most commercial systems require payment or external APIs. Lightweight, multi-domain systems that operate without user interaction data are rare in academic literature. REWIND addresses this by combining content-based filtering with optional LLM backends and deterministic fallback.

3. Methodology / Working

3.1 System Architecture

The system follows a modular three-stage pipeline:

• Data Layer: Loads PDF/text resume inputs and preprocesses them for analysis.
• Model Layer: Applies analyzer orchestration with backend selection (OpenAI → Groq → Local LLM → Free Heuristic).
• Service Layer: Resume parsing, skill extraction, scoring, upgrade generation, LaTeX compilation.
• Presentation Layer: FastAPI REST endpoints + web interface.

The analyzer orchestrator selects the best available backend based on configuration, normalizing all outputs to stable response schemas.

3.2 Core Components

Component	Role in Pipeline
FastAPI Routes	Handle CV scan, role analysis, resume upgrade, LaTeX compilation endpoints.
PDF Parser	Extract text from uploaded PDF and validate readability.
Analyzer Orchestrator	Select execution mode and normalize outputs into stable response schemas.
Free Analyzer	Perform deterministic skill extraction, matching, gap detection, and ATS suggestions.
Resume Template Service	Build upgraded resume snapshots and LaTeX output.

3.3 Machine Learning Component (scikit-learn)

Beyond heuristic analysis, the system integrates scikit-learn machine learning models for enhanced resume evaluation:

• TF-IDF Vectorizer: Converts resume and job descriptions into sparse numerical feature vectors, capturing word importance while diminishing common terms.
• Logistic Regression Classifier: Trained on 15+ synthetic resume samples (labeled as excellent/good/fair quality) to predict resume quality on scale 0-100.
• Cosine Similarity Matrix: Computes semantic similarity between resume and job description vectors, measuring role-fit percentage.
• Feature Extraction Pipeline: Analyzes word count, vocabulary richness, technical keyword density, and linguistic patterns.

All ML models are cached in memory on startup using Python pickle serialization, eliminating rebuild overhead. Predictions resolve in <50ms per query.

3.4 Core Scoring Logic

In free mode, score is derived from skill overlap and missing-skill penalty:

$$score = clamp(35 + 65 \cdot |M|/\max(|J|,1) - \min(2|G|,20), 20, 100)$$

where M is matched skills, J is job-description skills, and G is missing skills.

ML mode enhances scoring using Logistic Regression quality prediction and cosine similarity metrics from TF-IDF vectorization.

4. Results / Output

4.1 Model Performance

Category	Input	Top Result	Observation
CV Scan	PDF Upload	Score Card	Successfully scored with strengths/gaps identified.
Role Analysis	ML Engineer + JD	Match Score	Identified TensorFlow, PyTorch, MLOps gaps.
Resume Upgrade	Before Content	Improved Bullets	Generated +10-15 point ATS improvement.
Fuzzy Match	Atack on Titan	Attack on Titan	Spelling error correctly resolved.
Not Found	Unknown Title	—	Transparent honest error response.
LaTeX Export	Analysis Result	PDF Preview	Compiled resume ready for submission.

4.2 ML Results & Validation

ML Component	Test Input	Predicted Output
TF-IDF Quality Classifier	High-skill resume (Python, TensorFlow, MLOps, Docker, K8s)	Quality: 92/100 (Excellent) — Logistic Regression confidence: 92%
Cosine Similarity Matching	Resume vs ML Engineer JD	Semantic similarity: 0.77, Skill match: 34.8% — 18 matched, 10 missing
Feature Extraction NLP	Technical resume analysis	Tech keyword density: 8.2%, Vocabulary richness: 0.65, Detected keywords: 12
Fuzzy String Matching	Typos and variations	Attack-on-Titan → Attack on Titan, PyTorc → PyTorch (error-corrected)

All ML models (TF-IDF Vectorizer, Logistic Regression, Cosine Similarity) are trained once at startup using scikit-learn and persisted to disk via pickle serialization. Subsequent predictions use cached models for sub-50ms query response. The system demonstrates effective ML integration for resume quality prediction, semantic similarity analysis, and skill-gap identification without requiring external ML APIs.

5. Conclusion

This project delivers a practical AI-assisted toolkit for resume quality and role-fit analysis specifically tailored for technical roles. It combines API-driven intelligence with deterministic fallback logic, making it reliable across multiple deployment scenarios. The system successfully demonstrates: (1) multi-backend orchestration with graceful degradation (OpenAI → Groq → Ollama → Free), (2) role-specific skill gap analysis, (3) before-vs-after resume optimization, (4) production-ready LaTeX export, (5) accessible deployment without mandatory paid APIs, and (6) integrated machine learning models for quality prediction and semantic analysis. The ML component using scikit-learn (TF-IDF vectorization, Logistic Regression, Cosine Similarity) adds predictive intelligence without API dependencies. The solution is effective for students and early professionals seeking quick, structured, and ATS-aware career guidance.

6. Future Scope

- Add domain-specific scoring rubrics for ML, Data Science, Backend, Frontend, and DevOps roles.
- Integrate recruiter-specific templates and interview preparation guides.
- Build analytics dashboard for tracking score improvement across multiple resume iterations.
- Support multi-language resume analysis and localized job market guidance.
- Implement semantic embeddings (BERT) for nuanced query understanding.
- Add real-time collaboration features for team-based resume review.

7. References

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